Paradoxical Sex-Specific Patterns of Autoantibody Response to SARS-CoV-2 Infection

Yunxian Liu, PhD[1], Joseph E. Ebinger MD, MS[1], Sandy Joung, MHDS[1], Min Wu[1], Petra Budde, PhD[5], Jana Gajewski, MSc[5], Brian Walker, PhD[1], Rowann Mostafa, BS[1], Manuel Bräutigam, MSc[5], Franziska Hesping, BSc[5,6], Elena Schäfer, MSc[5], Ann-Sophie Schubert, MSc[5], Hans-Dieter Zucht, PhD[5], Jonathan Braun[3], Gil Y. Melmed, MD, MS[3], Kimia Sobhani, PhD[4], Moshe Arditi, MD[1,6], Jennifer E. Van Eyk, PhD[1,2] Susan Cheng, MD, MPH[1]* Justyna Fert-Bober, PhD[1,2]*

From Cedars-Sinai Medical Center, Los Angeles, California, USA.

Background

- Mechanisms underlying sex differences in both susceptibility and response to SARS-CoV-2 infection remain poorly understood.
- Despite classic autoimmune diseases being more prevalent in females, emerging data have revealed a paradoxical male predominance of autoimmune activation in the setting of severe COVID-19 illness.
- The extent to which such paradoxical sex differences in triggered autoimmunity may exist and persist across the broader that sex bias is unclear.
- We aimed to interrogate sex-specific autoimmune activation after SARS-CoV-2 exposure in the absence of severe clinical disease.

Methods

Study Sample

- Primary cohort: N=177 participants with confirmed prior COVID-19, based on presence of antibody to the nucleocapsid protein of SARS-CoV-2.
- Comparator sample: 53 pre-pandemic healthy controls (49% women, 51% men).

Serology

- We used a multiplex array to detect and quantify autoantibodies (AABs) to 91 antigens previously linked to a range of classic autoimmune diseases.

Analyses

- We examined the sex differences in the AAB responses to prior COVID-19.
- In sex-pooled and sex-specific analyses, we related the diversity of AAB responses with aggregate burden of symptoms as well as distinct individual symptoms attributed to prior COVID-19.
- We specifically used parametric tests, non-parametric tests, multivariable-adjusted analyses, clustering analyses, and co-occurrence analyses using R (v4.1.1) with statistical significance defined as a two-tailed P value <0.05.

Results

Figure 1. Sex-specific frequency of symptoms

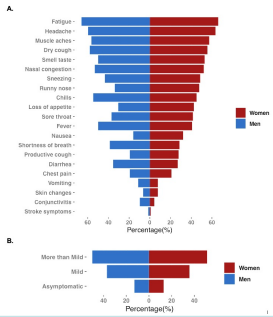


Figure 2. Sex Association with Autoantibody Response

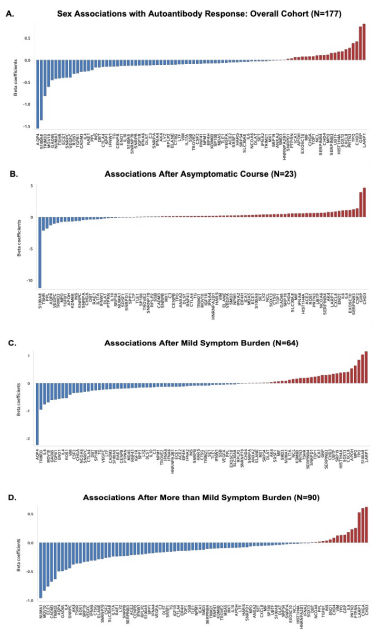
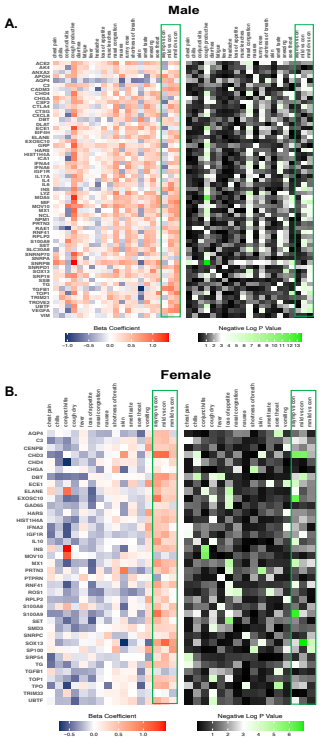


Figure 3. Autoantibody Associations with Symptoms



- The primary cohort of N=177 HCWs had a mean age of 35 [IQR: 30-44] years and included 65% women, 68% of non-white race, and 28% of Hispanic/Latinx ethnicity.
- Overall, males exhibited more diverse elevation in AAB levels than females, especially in relation to greater symptom burden.
- We observed significant sex-specific patterns of autoreactivity linked to the presence, timing, and clustering of symptoms associated with prior COVID-19 illness.
- Whereas the overall AAB response was more prominent in women following asymptomatic infection, the breadth and extent of AAB reactivity was more prominent in men following at least mildly symptomatic infection.
- The observed reactivity included distinct antigens with molecular homology with SARS-CoV-2.

Conclusions

In summary, this comprehensive study of AABs to a wide array of antigens found that male sex carries the risk of diverse autoimmune activation following symptomatic COVID-19 illness, whereas female sex carries risk for a distinct profile of autoimmune activation following asymptomatic SARS-CoV-2 exposure. Importantly, both sets of sex specific AAB reactivity patterns were found to persist up to 6 months following associated symptomatology. Further understanding the nature of triggered and persistent AAB activation among individuals who are exposed to SARS-CoV-2 – and vulnerable to its potentially morbid clinical sequelae – will be essential for developing effective interventions and therapeutics.

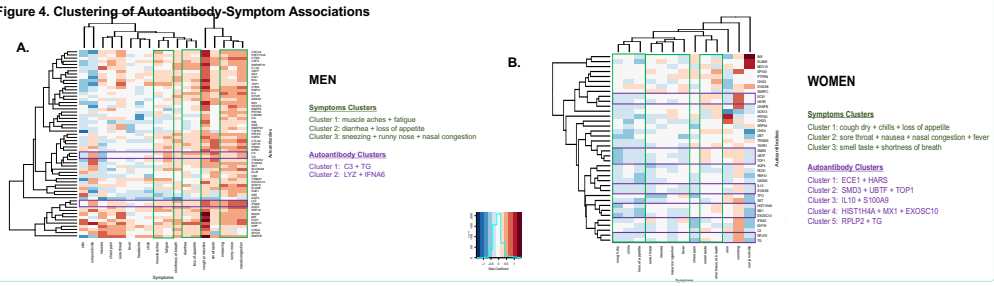



Figure 4. Clustering of Autoantibody-Symptom Associations